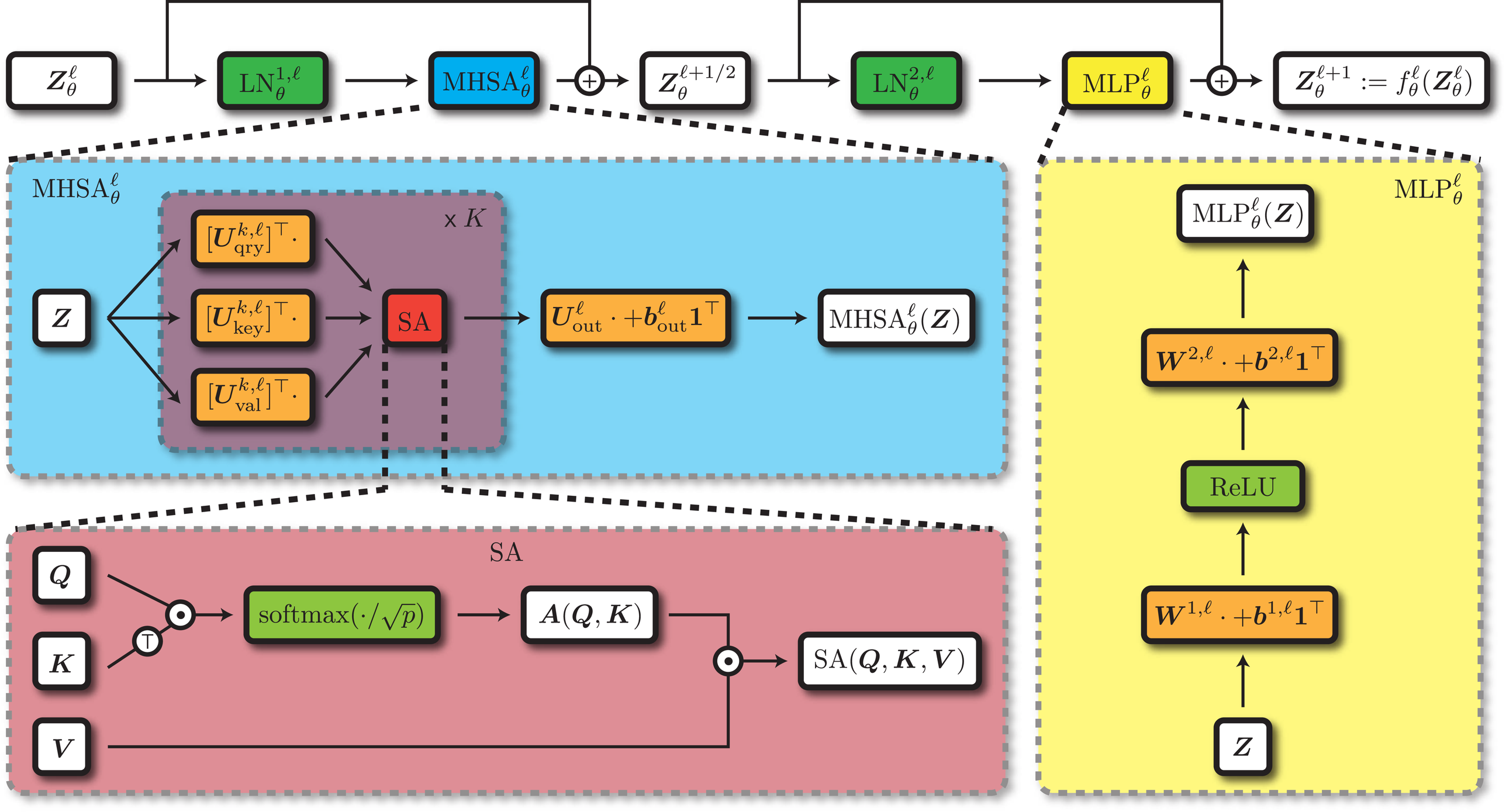
Z_θ^ℓ
$\mathrm{LN}_\theta^{1,\ell}$
$\mathrm{MHSA}_\theta^\ell$
$Z_\theta^{\ell+1/2}$
$\mathrm{LN}_\theta^{2,\ell}$
MLP_θ^ℓ
$Z_\theta^{\ell+1} := f_\theta^\ell(Z_\theta^\ell)$
$\mathrm{MHSA}_\theta^\ell$
$\times K$
$[U_{\mathrm{qry}}^{k,\ell}]^\top \cdot$
$[U_{\mathrm{key}}^{k,\ell}]^\top \cdot$
$[U_{\mathrm{val}}^{k,\ell}]^\top \cdot$
Z
SA
$U_{\mathrm{out}}^\ell \cdot + b_{\mathrm{out}}^\ell \mathbf{1}^\top$
$\mathrm{MHSA}_\theta^\ell(Z)$
SA
Q
K
V
$\top$
$\mathrm{softmax}(\cdot/\sqrt{p})$
$A(Q,K)$
$\mathrm{SA}(Q,K,V)$
MLP_θ^ℓ
$\mathrm{MLP}_\theta^\ell(Z)$
$W^{2,\ell} \cdot + b^{2,\ell}\mathbf{1}^\top$
ReLU
$W^{1,\ell} \cdot + b^{1,\ell}\mathbf{1}^\top$
Z